Calculus II

Worksheet

1. If the amount of mayhem is constant in an episode of the Muppet Show, there is a simple formula that gives the amount of mess created:

$$\text{Mess} = \text{Mayhem} * \text{Time}$$

That is, the mess is the product of the mayhem and the length of time it lasts. Typically, however, the mayhem varies over time. In a certain 30 minute episode the amount of mayhem was given by the function $M(t) = 4\ln(t+1)$, where $0 \le t \le 30$. Describe in complete detail how integration can be used to find the total mess in the episode.

2. It is a well-known fact that the anger generated by a fan during a football game is caused by both the frustration they experience and the length of time it occurs. The anger produced by a constant ammount of frustration is given by the formula

$$\text{Anger} = \frac{1}{3}(\text{Frustration})^2 * \text{Time}.$$

In the first quarter of a game, the Frustration was given by $F(t) = \sin\left(\frac{\pi}{30}t\right)$, where $0 \le t \le 15$.

Describe in complete detail how integration can be used to find the total Anger in the first quarter. Discuss the role played by Riemann sums, and set up an integral to find this Anger.

3. A conical tank has a radius of 5 ft and a height of 13 ft. Set up an integral to determine how much work is done in emptying it if
   (a) It is filled with milk (which weighs 64.5 lb/ft^3).

   (b) It is filled to a depth of 9 ft and the milk must be pumped out a 2 ft pipe at the top of the tank.

4. A dam has a triangular gate (triangle pointing upwards) with base 4 ft and height 5 ft. The top of the gate (the point of the triangle) is 3 feet below the water level. Set up an integral to find the force on the gate. (Recall the density of water is 62.4 lb/ft^3.)

5. Set up integrals to find the mass, moment about the x-axis, and moment about the y-axis of the lamina bounded by $y = \sqrt{x}$ and $y = \frac{1}{3}x$. Give the formula to find the coordinates of the center of mass.

6. A spring has a natural length of 5 ft. A force of 4 lb stretches this spring 1/4 ft.
   (a) Find the work done in stretching this spring from its natural length to 7 ft.

   (b) find the work done in stretching this spring from 8 ft to 9 ft.

7. How much work is done in emptying a cylindrical tank of radius 5 ft and height 10 ft by pumping the the water through a pipe extending 4 ft above the tank. (Hint: Water weighs 62.4 lb/ft^3.)

8. A rectangular gate built into a dam is 4 ft high and 8 ft wide. The top of the gate is 5 ft below the surface of the water. What is the force on the gate?

9. What would be the force if the gate were the upper half of disc of radius 4 (and the top of the gate is 5 ft below the surface of the water)?

10. Find the mass, moments, and center of mass of a the triangular lamina bounded by the x- and y- axes and $y = 4 - 2x$. (As always, assume the density of this lamina to be 1.)